Zhixiong Xiao

Columbus, OH | xiao.1314@osu.edu | +1 3808952174 | xiaozxiong.github.io

EDUCATION

The Ohio State University, Ph.D. in Computer Science and Engineering — Sep. 2026 – Jun. 2031

Shandong University, M.Eng. in Computer Science and Technology — Sep. 2022 – Jun. 2025

- GPA: 4.02/5.00

Dalian Maritime University, B.Eng. in Computer Science and Technology — Sep. 2018 – Jun. 2022

- GPA: 3.96/5.0

EXPERIENCE

Research Assistant, School of Computer Science and Technology, Shandong University — Sep. 2025 – Jul. 2026

PUBLICATIONS

GAS: An I/O-Efficient GPU-centric ANNS System towards Billion-Scale Datasets — Jul. 2026

Zhengrui Su, Hao Huang, Mengbai Xiao, ***Zhixiong Xiao***, Zhaoyan Shen, Dongxiao Yu, Zhaoqi Zhou, Guoliang Li
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)

PARS: Optimizing High-Dimensional Vector Search with Dimensional Reduction and Ray Tracing Core — Jan. 2026

Yiling Ma, Mengbai Xiao, ***Zhixiong Xiao***, Yuan Yuan, Dongxiao Yu, Guanghui Zhang, Feng Li
46th IEEE International Conference on Distributed Computing Systems (ICDCS), 2026

A Case Study for Ray Tracing Cores: Performance Insights with Breadth-First Search and Triangle Counting in Graphs — Jan. 2025

Zhixiong Xiao, Mengbai Xiao, Yuan Yuan, Dongxiao Yu, Rubao Lee, Xiaodong Zhang
International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS), 2025

Perceptually Driven Color Ordering for Palette Design — Jul. 2022

Zhixiong Xiao, Wensong Xu, Qiong Zeng
Journal of Computer-Aided Design & Computer Graphics

SCHOLARSHIPS

- Third-Class Academic Scholarship, Shandong University — Oct. 2024
- First-Class Academic Scholarship, Shandong University — Nov. 2023
- Weichai Power Scholarship, Shandong University — Oct. 2023
- Second-Class Academic Scholarship for New Students, Shandong University — Oct. 2022
- Competition Scholarship, Dalian Maritime University — Dec. 2020
- National Encouragement Scholarship, Dalian Maritime University — Nov. 2019

AWARDS

- Outstanding Graduate of Dalian Maritime University — May 2022
- Bronze Medal, The 45th ACM International Collegiate Programming Contest (ICPC), Shenyang Site — Jul. 2021
- Second Prize, C/C++ Group, The 12th Lanqiao Cup National Software and Information Technology Talent Competition, National Finals — Jun. 2021
- Bronze Medal, The 45th ACM International Collegiate Programming Contest (ICPC), Yinchuan Site — May 2021
- Third Prize (Individual), The 6th China Collegiate Computing Contest - Group Programming Ladder Tournament, National Final — May 2021

- Silver Medal, The 5th China Collegiate Computing Contest - Group Programming Ladder Tournament, National Final — Dec. 2020
- Bronze Medal, The 6th China Collegiate Programming Contest (CCPC), Changchun Site — Nov. 2020
- Third Prize, The 14th Northeastern China Collegiate Programming Contest — Oct. 2020
- Third Prize of Liaoning Provincial National Undergraduate Mathematical Contest in Modeling — Oct. 2019
- Third Prize, Dalian Mathematics Competition — Jun. 2019

INTERESTS AND SKILLS

Interests: Computer Systems, High-Performance Computing, GPU Architecture, Vector Database, ML Systems, Agents

Programming Skills: C/C++, CUDA, Python, Triton, OpenMP, OptiX, Torch, Git

TEACHING EXPERIENCE

Teaching Assistant at Shandong University

- Data Structures and Algorithms, Instructor: Mengbai Xiao — Fall 2024
- Data Structures and Algorithms, Instructor: Mengbai Xiao — Fall 2023
- Curriculum Design for Operating Systems, Instructor: Qiang Yang — Spring 2023